

LEDGER · 2026-08-29 · NANOMATERIALS

High-Photostability Anti-Counterfeiting Inks via Hydrothermal Carbonization of Biomass into Carbon Quantum Dots

PRODUCT



A product dossier of The Absence Audit. The current audit flags this concept — Margin fails the operating-cost check — and the full disclosure ships inside the dossier.

Incumbent benchmark

Shenzhen Goosam UV Fluorescent Water-Based Invisible Ink (L3258/L3151 series)

Entry application

B2B secondary packaging converters using Epson Piezoelectric inkjet printheads on porous cardboard and paper

Measured advantage	Retains >90% signal after 55 hours continuous UV exposure versus incumbent losing >70% signal in a few hours (>10x operational photostability improvement)
Time to first revenue	4 months
Gross margin at parity	52.5%
Startup CapEx	\$45,690
Payback from first sale	12.0 months
Capital productivity	3.55x
Regulatory risk	Low — no material regulatory CapEx reported (\$0 estimated)

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Mechanism, operating protocol with real conditions and yields, computed unit economics with a six-scenario stress test, the absence audit, and every resolved citation.

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Assessed 2026-08-29 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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